

Cost of Capital

Creation of Value

- If the return on a project exceeds what the financial markets require, it is said to earn an *excess return*. This excess return, as we define it, represents the creation of value.
- Simply put, the project earns more than its economic keep. Finding and undertaking these value-creating (positive NPV) projects increases a company's common stock share price.
- Value creation has several sources, but perhaps the most important are industry attractiveness and competitive advantage. These are the things that give rise to projects with positive net present values – ones that provide expected returns in excess of what the financialmarkets require.

Creation of Value

- **Industry Attractiveness**

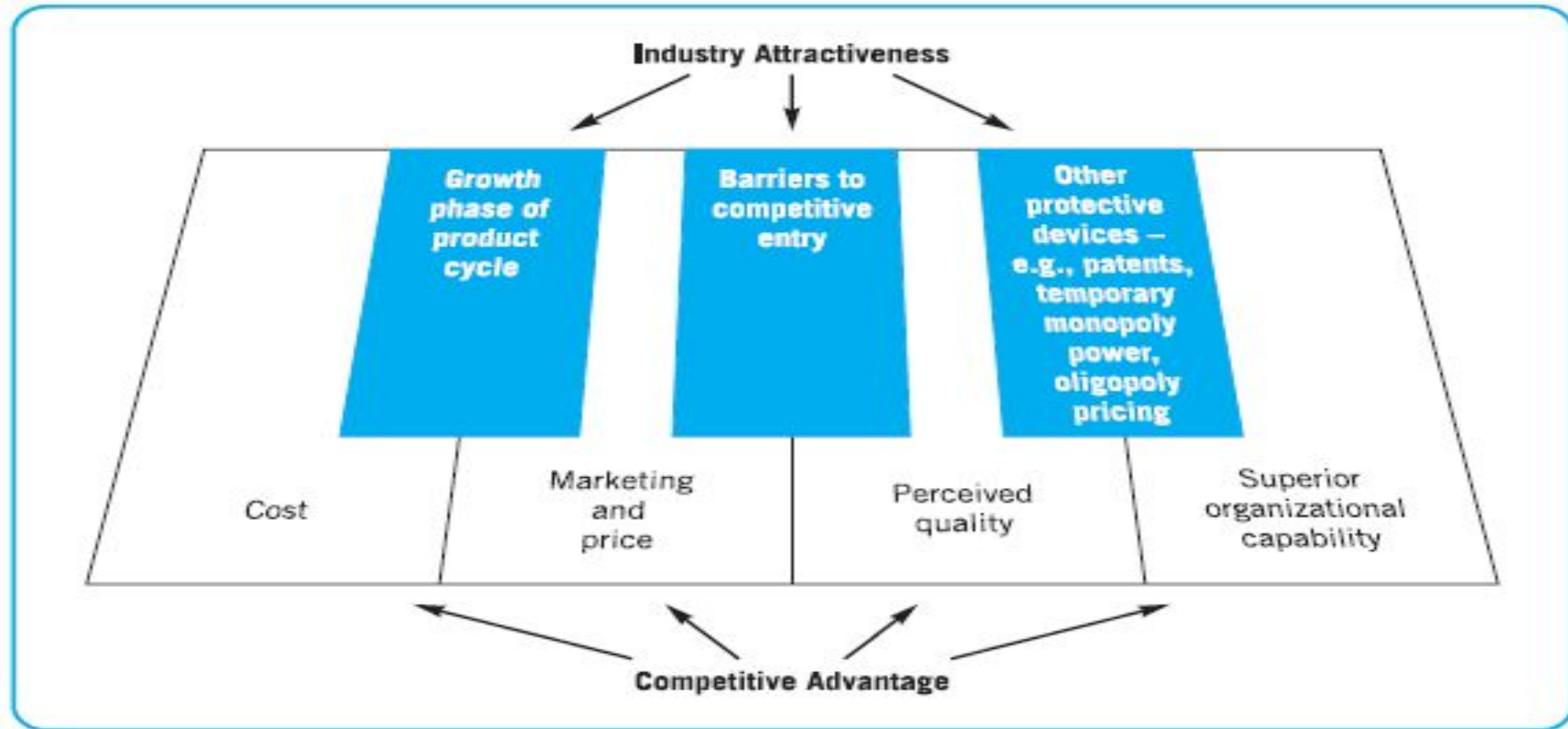
- Favorable industry characteristics include positioning in the growth phase of a product cycle, barriers to competitive entry, and other protective devices, such as patents, temporary monopoly power, and/or oligopoly pricing where nearly all competitors are profitable.
- In short, industry attractiveness has to do with the relative position of an industry in the spectrum of value-creating investment opportunities.

Creation of Value

- **Competitive Advantage**

- Competitive advantage involves a company's relative position within an industry.
- The company could be multidivisional, in which case competitive advantage needs to be judged industry by industry. The avenues to competitive advantage are several: cost advantage, marketing and price advantage, perceived quality advantage, and superior organizational capability (corporate culture).
- Competitive advantage is eroded with competition. Relative cost, quality, or marketing superiority, for example, are conspicuous and will be attacked.
- A successful company is one that continually identifies and exploits opportunities for excess returns. Only with a sequence of short-run advantages can any overall competitive advantage be sustained.

Creation of Value



Overall Cost of Capital of the Firm

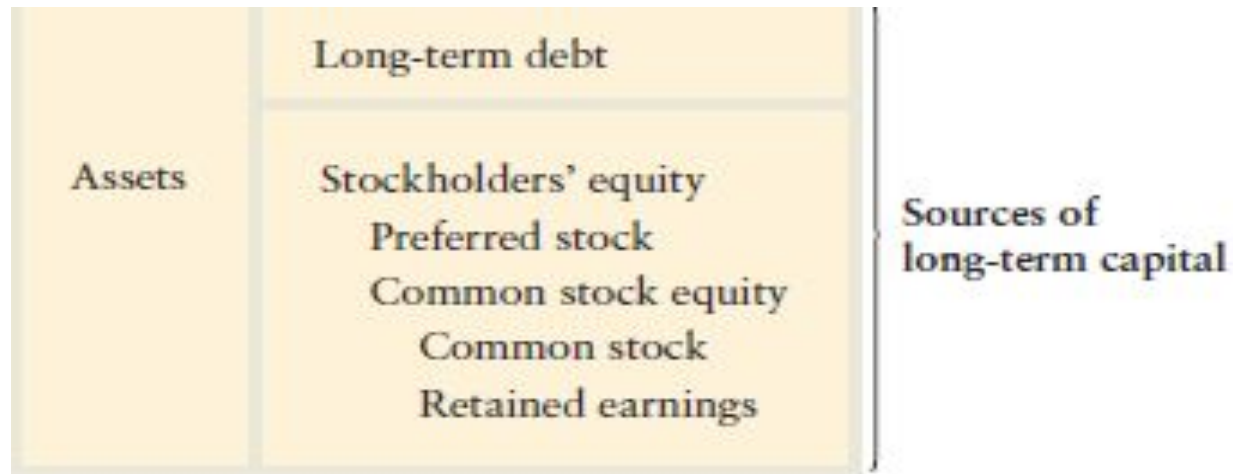
- The **cost of capital** represents the firm's cost of financing and is the minimum rate of return that a project must earn to increase firm value.
- When investors (i.e., stockholders and lenders) provide funding to a firm, they do so with the expectation that they will earn a return. The returns that investors require constitute a cost of capital the firm must pay to investors.
- Investments that produce returns above the cost of capital will exceed investors' expectations and increase the value of the firm. In contrast, projects with returns below the cost of capital will decrease firm value.

Capital structure

- Firms need money to pay for their investments. The term capital refers to a firm's long-term sources of financing, which include debt and equity.
- A firm's **capital structure** refers to the mixture of debt and equity financing it employs. Some firms prefer capital structures that are close to 100% equity, while other firms borrow heavily.
- The weighted average cost of capital (WACC) is a weighted average of the firm's cost of debt and equity capital, where the weights correspond to the percentage of each type of financing used by the firm.
- The WACC reflects the firm's overall cost of financing, not just the cost of one financing source.
- Managers must take into account the costs of both forms of capital when they make investment decisions because the cash flows produced by investments must be sufficient to satisfy the expectations of both types (debt and equity) of investors.

Sources Of Long-term Capital

- Long-term capital for firms derives from four basic sources: long-term debt, preferred stock, common stock, and retained earnings. All entries on the right-hand side of the balance sheet, other than current liabilities



Long-term Capital and required rate of return

- The overall cost of capital of a firm is a proportionate average of the costs of the various components of the firm's financing.
- **Cost of equity capital** The required rate of return on investment of the common shareholders of the company.
- **Cost of debt (capital)** The required rate of return on investment of the lenders of a company.
- **Cost of preferred stock (capital)** The required rate of return on investment of the preferred shareholders of the company
- All costs will be expressed on an after-tax basis, to conform to the expression of investment project cash flows on an after-tax basis.

Cost of Long-Term Debt

- Firms that borrow money pay interest to lenders. That interest is a cost to the borrower, but interest alone does not fully reflect a firm's borrowing cost. If a firm borrows money by issuing bonds, it will incur costs specifically tied to the process of issuing new securities, separate from the interest payments made on those securities.
- The **before-tax cost of longterm debt**, rd , is the before-tax financing cost associated with new funds raised through long-term borrowing, taking into account the firm's interest payments and other borrowing costs.
- Typically, firms raise long-term debt by issuing corporat bonds, though they may also borrow on a long-term basis from banks or other financial institutions.

Cost of Long-Term Debt

- The **net proceeds** from the sale of a bond, or any security, are the funds that the firm receives from the sale. Net proceeds are less than total proceeds due to **flotation costs**, which represent the total costs of issuing and selling securities.
- These costs apply to all public offerings of securities: debt, preferred stock, and common stock. They include two components: (1) *underwriting costs*, or compensation earned by investment bankers for selling the security; and (2) *administrative costs*, or issuer expenses such as legal and accounting costs.

Cost of Long-Term Debt

- The explicit cost of debt can be derived by solving for the discount rate, k_d or r_d , that equates the market price of the debt issue with the present value of interest plus principal payments and by then adjusting the explicit cost obtained for the tax deductibility of interest payments.
- The discount rate, k_d , known as the *yield to maturity*, is solved for by making use of the formula where P_0 is the current market price of the debt issue; $\sum$ denotes the summation for periods 1 through n , the final maturity; I_t is the interest payment in period t ; and P_t is the payment of principal in period t . If principal payments occur only at final maturity, only P_n will occur.

$$P_0 = \sum_{t=1}^n \frac{I_t + P_t}{(1 + k_d)^t}$$

Cost of Long-Term Debt

After-tax Cost Of Debt

- Unlike dividends on stock, interest payments on bonds are tax deductible, so the interest expense on debt reduces taxable income and, therefore, the firm's tax liability. In effect, this means the firm's after-tax cost of debt is less than the return paid to bondholders

$$k_i = k_d(1 - t)$$

$$\text{After-tax cost of debt} = r_d \times (1 - T)$$

Example

- Duchess receives proceeds of \$980 by issuing a 20-year bond with a \$1,000 par value and 6% coupon interest rate. To calculate the before-tax cost of debt, begin by writing down the cash flows associated with this bond issue. The cash flow pattern consists of an initial inflow (the net proceeds) followed by a series of annual outflows (the interest payments). In the final year, when the debt is retired, an outflow representing the repayment of the principal also occurs. The cash flows associated with Duchess Corporation's bond issue are as follows:

<u>End of year(s)</u>	<u>Cash flow</u>
0	\$ 980
1-20	-60
20	-1,000

Example

- The before- tax cost of debt (yield to maturity) equals 6.177%.
- Duchess Corporation pays a 21% tax rate. Using the 6.177% before-tax debt cost calculated above and applying Equation 9.1, we find a after-tax cost of debt of
- $4.88 = [6.177 \% \times (1 - 0.212)]$

Example

- Kait and Kasim Sullivan, a married couple in the 28% income-tax bracket, wish to borrow \$60,000 to pay for improvements to their home. To finance the purchase, either they can borrow from a relative at an annual interest rate of 4.5%, or they can take a second mortgage on their home. The best annual rate they can get on the second mortgage is 5.5%.
- If they borrow from the relative, the interest will not be deductible for federal tax purposes. However, the interest on the second mortgage would be deductible because the tax law allows individuals to deduct home mortgage interest. To choose the least-cost financing, the Sullivans calculate the after-tax cost of both loans. Because interest on the loan from a relative is not deductible, its after-tax cost equals its before-tax cost of 4.5%.

Example

$$\text{After-tax cost of debt} = 5.5\% \times (1 - 0.28) = 5.5\% \times 0.72 = 3.96\%$$

Because the 3.96% after-tax cost of the second mortgage is less than 4.5%, the Sullivans may decide to use the second mortgage to finance the home improvement project.

Cost of Preferred Stock

- **Cost of Preferred Stock** is the required rate of return on investment of the preferred shareholders of the company.
- Preferred stock represents a special type of ownership interest in the firm. It gives preferred stockholders the right to receive dividends before the firm can distribute any earnings to common stockholders.
- When companies issue preferred shares, the shares usually pay a fixed dividend and have a fixed par value. When discussing the features of a particular preferred stock issue, investors may refer to the fixed dividend either in dollar terms or as a percentage of the stock's par value

Cost of Preferred Stock

- The **cost of preferred stock, r_p** , is the ratio of the preferred stock dividend to the firm's net proceeds from the sale of the preferred stock. As with bonds, net proceeds equal the total proceeds minus any flotation costs. The following equation gives the cost of preferred stock, r_p , in terms of the annual dollar dividend, D_p , and the net proceeds from the sale of the stock, N_p :

$$r_p = \frac{D_p}{N_p}$$

$$k_p = D_p / P_0$$

Example

- Duchess Corporation is contemplating issuance of an 8% preferred stock they expect to sell at par value for \$80 per share. The cost of issuing and selling the stock will be \$2.50 per share.

Example

- The first step in finding the cost of the stock is to calculate the dollar amount of the annual preferred dividend, which is

$$\$6.40 (0.08 \times \$80).$$

- The net proceeds per share from the proposed sale of stock equals the sale price minus the flotation costs

$$(\$80 - \$2.50 = \$77.50).$$

- Substituting the annual dividend, D_p , of \$6.40 and the net proceeds, N_p , of \$77.50 gives the cost of preferred stock,

$$, 8.258\% (\$6.4 \div \$77.50).$$

Cost of Common Stock/Equity

- The **cost of common stock equity** reflects the costs that firms incur to utilize common stock financing. Just as the costs of debt and preferred stock were influenced by the required returns of bondholders and preferred stockholders, the cost of common stock equity will reflect the return required by the firm's stockholders.
- However, calculating the required return for common stock is more difficult than finding the required return on bonds or preferred shares.
- Moreover, for most common stocks, a significant component of the return that investors receive comes in the form of a capital gain resulting from reinvestment of the firm's earnings in the business. Therefore, managers require a different approach to calculate the required return on common stock. Three techniques are available.
 - Dividend Discount Model
 - Capital-Asset Pricing Model
 - Before-Tax Cost of Debt plus Risk Premium

Cost of Common Stock/Equity

- The **cost of equity capital**, k_e , is the discount rate that equates the present value of all expected future dividends with the current market price of the stock.
- Dividend Discount Model / Gordon Growth Model

$$P_0 = \frac{D_1}{r_s - g}$$

$$r_s = \frac{D_1}{P_0} + g$$

where

P_0 = current value of common stock

D_1 = dividend expected in one year

r_s = required return on common stock

g = constant rate of growth in dividends

Example

- Duchess Corporation wishes to determine the required return, r_s , on its common stock. The market price, P_0 , is \$50 per share. Duchess recently paid a \$3.80 dividend.
- The company has increased its dividend for several consecutive years. Just five years ago, Duchess paid a dividend of \$2.98 on its common stock.

Example

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$$r_s = \frac{\$3.99}{\$50} + 0.05 = 0.0798 + 0.05 = 0.1298 = 12.98\%$$

Example

- For example, if dividends are expected to grow at an 8 percent annual rate into the foreseeable future, the constant growth model presented in Chapter 4 might be used to determine the required rate of return.
- If the expected dividend in the first year were \$2 and the present market price were \$27, we would have

$$\begin{aligned}k_e &= (D_1/P_0) + g \\ &= (\$2/\$27) + 0.08 = \mathbf{15.4\%}\end{aligned}$$

Cost of Common Stock/Equity

- *Growth Phases Model*
- The growth phases assumption leads to the following formula (assume 3 growth phases):

$$P_0 = \sum_{t=1}^a \frac{D_0(1 + g_1)^t}{(1 + k_e)^t} + \sum_{t=a+1}^b \frac{D_a(1 + g_2)^{t-a}}{(1 + k_e)^t} + \sum_{t=b+1}^{\infty} \frac{D_b(1 + g_3)^{t-b}}{(1 + k_e)^t}$$

Cost of Equity: Capital-Asset Pricing Model Approach

- Capital asset pricing model (CAPM) describes the relationship between the required or expected return on some asset j , r_j , and the non-diversifiable risk of the asset as measured by the beta coefficient, β_j .
- The CAPM says that

$$r_j = R_F + [\beta_j \times (r_m - R_F)] \quad (9.5)$$

where

R_F = risk-free rate of return

r_m = expected market return; expected return on the market portfolio

Example

- Duchess Corporation now wishes to calculate the required return on its common stock, r_s , by using the CAPM. The firm's investment advisors and its own analysts indicate that the risk-free rate, R_F , equals 3%; the firm's beta, b , equals 1.5; and the expected market return, r_m , equals 9%. Substituting these values into

$$r_s = 3.0\% + [1.5 \times (9.0\% - 3.0\%)] = 3.0\% + 9\% = 12.0\%$$

Cost of Equity: Before-Tax Cost of Debt Plus Risk Premium Approach

- The **cost of equity capital**, k_e , is the sum of the before-tax cost of debt and a risk premium in expected return for common stock over debt.

$$k_e = \begin{array}{l} \text{Before-tax} \\ \text{cost of} \\ \text{debt } (k_d) \end{array} + \begin{array}{l} \text{Risk premium in} \\ \text{expected return for} \\ \text{stock over debt} \end{array}$$

- Assume that *Basket Wonders (BW)* typically adds a 2.75% premium to the before-tax cost of debt.

$$k_e = k_d + \text{Risk Premium}$$

$$= 10\% + 2.75\%$$

$$k_e = 12.75\%$$

Cost of Common Stock/Equity

- **Flotation Costs and the Cost of Common Equity**

- Debt and preferred stock, the firm may face flotation costs, which increase the cost of capital to the firm. Thus far we have ignored these costs related to common equity.
- Unlike the CAPM, the constant-growth model offers a relatively easy way to adjust for flotation costs in finding the cost of common stock.
- When a firm issues new common stock, a variety of costs reduce the net proceeds the firm receives. These costs include the underwriting fees and administrative costs.
- In addition, investment bankers usually sell new shares of common stock at a slight discount to their current market value, a phenomenon known as *underpricing*. Underpricing represents a cost to issuers because it requires them to sell a greater number of shares (which over time will receive a greater amount of dividends) than would be the case if shares were sold at full market value.

Cost of Common Stock/Equity

- The **cost of a new issue of common stock, r_n** , is determined by calculating the cost of common stock, net of underpricing and associated flotation costs. We can use the constant-growth valuation model expression for the required return on common stock, r_s , as a starting point.

$$r_n = \frac{D_1}{N_n} + g$$

Example

- In the constant-growth valuation example, we found that Duchess Corporation's required return on common stock, r_s , was 13%, using the following values: an expected dividend, D_1 , of \$3.99; a current market price, P_0 , of \$50; and an expected growth rate of dividends, g , of 5%.
- To determine its cost of new common stock, r_n , Duchess Corporation estimates that new shares will sell for \$48. Thus, Duchess's shares will be underpriced by \$2 per share. A second cost associated with a new issue is flotation costs of \$1.50 per share that would be paid to issue and sell the new shares.

Example

Subtracting the \$3.50-per-share underpricing and flotation cost from the current \$50 share price results in expected net proceeds of \$46.50 per share. Substituting $D_1 = \$3.99$, $N_n = \$46.50$, and $g = 5\%$ into Equation 9.6 results in a cost of new common stock, r_n :

$$r_n = \frac{\$3.99}{\$46.50} + 0.05 = 0.0858 + 0.05 = 0.1358 = 13.58\%$$

Duchess Corporation's cost of new common stock is therefore between 13% and 14%.

Weighted Average Cost of Capital

- The *weighted average cost of capital (WACC)*, r_{wacc} , reflects the expected average cost of the different forms of capital used by a company.
- It equals the weighted average cost of each specific type of capital, where the weights equal the proportion of each capital source in the firm's capital structure.

$$r_{wacc} = (w_d \times r_d)(1 - T) + (w_p \times r_p) + (w_s \times r_{s\text{ or }n}) \quad (9.8)$$

where

w_d = proportion of long-term debt in capital structure

w_p = proportion of preferred stock in capital structure

w_s = proportion of common stock equity in capital structure

$w_d + w_p + w_s = 1.0$

Weighted Average Cost of Capital

- We want to highlight four important points related to Equation 9.8:

1. The weights must be nonnegative and sum to 1.0, so that the WACC accounts for all financing costs in the firm's capital structure.
2. The weights are based on the market value of each capital source as a percentage of the market value of the firm's total capital. For example, w_s is based on the market value of the firm's stock, not on its par value or the value of equity shown on the firm's balance sheet.
3. We multiply the firm's common stock equity weight, w_s , by either the required return on the firm's stock, r_s , or the cost of new common stock, r_n .

Remember that when firms finance investments with retained earnings, the cost of retained earnings, rr , is the same as the required return on the firm's stock. However, when the firm issues new shares, the cost of equity is higher due to flotation costs, so the firm may use r_n in Equation 9.8.

4. We multiply the firm's cost of debt by $(1 - T)$ to capture the tax deduction tied to interest payments.

Weighted Average Cost of Capital Example

- In earlier examples, we found that the costs of the various types of capital for Duchess Corporation were:

$$r_d(1 - T) = 4.880\% = 4.88\%$$

$$r_p = 8.258\% = 8.26\%$$

$$r_s = 13.00\%$$

- Duchess has total capital with a market value of \$1 billion. The market values of the firm's outstanding long-term debt, preferred stock, and common stock are \$400 million, \$100 million, and \$500 million, respectively.

Weighted Average Cost of Capital Example

TABLE 9.2

Calculation of the Weighted Average Cost of Capital for Duchess Corporation

Source of capital	Weight w	Cost r	Weighted cost $w \times r$
Long-term debt	0.40	4.88%	1.95%
Preferred stock	0.10	8.26	0.83
Common stock equity	<u>0.50</u>	13.00	<u>6.50</u>
Totals	1.00		WACC = <u>9.28%</u>

Weighted Average Cost of Capital Example

	AMOUNT OF FINANCING	COST
Debt	\$ 30 million	6.6%
Preferred stock	10 million	10.2
Common stock equity	60 million	14.0
	<u>\$100 million</u>	

	(1)	(2)	(1) × (2)
	COST	PROPORTION OF TOTAL FINANCING	WEIGHTED COST
Debt	6.6%	30%	1.98%
Preferred stock	10.2	10	1.02
Common stock equity	14.0	60	8.40
		<u>100%</u>	<u>11.40%</u>

Capital Structure Weights

- WACC uses **market value weights** for the percentages of each type of financing, rather than book or par values. This is because market value better reflects the value of the funds that investors have placed in the hands of managers and on which investors expect to earn a return.
- **Target capital structure** The mix of debt and equity financing that a firm desires over the long term. The target capital structure should reflect the optimal mix of debt and equity for a particular firm.

Limitations of WACC

- With the calculation of a weighted average cost of capital, the critical question is whether the figure represents the firm's real cost of capital. The answer depends on how accurately we have measured the individual marginal costs, on the weighting system, and on certain other assumptions.

Limitations of WACC

- **Weighting System.** The critical assumption in any weighting system is that the firm will in fact raise capital in the proportions specified. Because the firm raises capital *marginally* to make *marginal* investments in new projects, we need to work with the marginal cost of capital for the firm as a whole.
- This rate depends on the package of funds employed to finance investment projects. In other words, our concern is with new or incremental capital, not with capital raised in the past. For the weighted average cost of capital to represent a marginal cost, the weights employed must be marginal.
- The real weighted average cost of capital will differ from that calculated and used for capital investment decisions. If the real cost is greater than that which is measured, certain investment projects that will leave investors worse off than before will be accepted. On the other hand, if the real cost is less than the measured cost, projects that could increase shareholder wealth will be rejected.

Limitations of WACC

- **Flotation costs** involved in the sale of debt instruments, preferred stock, or common stock affect the profitability of a firm's investments. In many cases, the new issue must be priced below the market price of existing financing.
- In addition, there are out-of-pocket flotation costs. Owing to flotation costs, the amount of funds the firm receives is less than the price at which the issue is sold. The presence of flotation costs in financing requires that an adjustment be made in the evaluation of investment proposals.

Limitations of WACC

- One approach, which we refer to as the *adjustment to initial outlay* (AIO) method, treats the flotation costs of financing as an addition to the initial cash outlay for the project.
- According to this procedure, the net present value of a project is computed according to

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1+k)^t} - (ICO + \text{Flotation costs})$$

- where CF_t is the project cash flow at time t , ICO is the initial cash outlay required for the project, and k is the firm's cost of capital.

Limitations of WACC

- A second, more traditional approach calls for an upward adjustment of the cost of capital when flotation costs are present. This method, which we refer to as the *adjustment to discount rate* (ADR) procedure, thus adjusts a project's discount rate for flotation costs and not the project's cash flows.
- Under this procedure each component cost of capital would be recalculated by finding the discount rate that equates the present value of cash flows to the suppliers of capital with the *net proceeds* of a security issue, rather than with the security's market price.
- The resulting “adjusted” component costs would then be weighted and combined to produce an overall “adjusted” cost of capital for the firm.

Rationale for a Weighted Average Cost

- The rationale behind the use of a weighted average cost of capital is that, by financing in the proportions specified and accepting projects yielding more than the weighted average required return, the firm is able to increase the market price of its stock.
- This increase occurs because investment projects are expected to return more on their equity-financed portions than the required return on equity capital, k_e . Once these expectations are apparent to the marketplace, the market price of the firm's stock should rise, all other things remaining equal.
- The firm has accepted projects that are expected to provide a return greater than that required by investors at the margin, based on the risk involved.

Economic Value Added (EVA)

- **Economic value added (EVA)** A measure of business performance. It is a type of economic profit, which is equal to a company's after-tax net operating profit minus a dollar cost of capital charge (and possibly including some adjustments).
- The EVA approach gives managers a tool to measure an investment's performance on a year-by-year basis as well as over its entire life.
- To calculate EVA, each period starts with a project's net operating profit after tax (NOPAT), which is equal to the project's operating cash flow (OCF) minus any depreciation. Next, subtract an EVA capital charge that represents the return that the firm's investors demand on the capital invested in the project that period.
- The amount of capital invested in a project can change from one year to the next due to depreciation as well as any new investments in fixed assets or working capital required to keep the project going.

$$\begin{aligned} \text{EVA} &= \text{NOPAT} - \text{EVA charge} \\ &= (\text{OCF} - \text{Depreciation}) - (\text{Invested capital} \times \text{cost of capital}) \end{aligned} \quad (10.3)$$

Economic Value Added (EVA)

- The EVA method determines whether a project earns a *pure economic profit*.
- When accountants say that a firm has earned a profit, they mean that revenues are greater than expenses.
- But the term **pure economic profit** refers to a profit that is higher than expected given the competitive rate of return on a particular line of business. A firm that shows a positive profit on its income statement may or may not earn a pure economic profit, depending on how large the profit is relative to the capital invested in the business.

Economic Value Added (EVA)

Project A				
Year	OCF	Depreciation	NOPAT	Beginning of Year Invested Capital
1	\$140,000	\$84,000	$\$140,000 - \$84,000 = \$56,000$	$\$420,000 - \$84,000 \times 0 = \$420,000$
2	\$140,000	\$84,000	$\$140,000 - \$84,000 = \$56,000$	$\$420,000 - \$84,000 \times 1 = \$336,000$
3	\$140,000	\$84,000	$\$140,000 - \$84,000 = \$56,000$	$\$420,000 - \$84,000 \times 2 = \$252,000$
4	\$140,000	\$84,000	$\$140,000 - \$84,000 = \$56,000$	$\$420,000 - \$84,000 \times 3 = \$168,000$
5	\$140,000	\$84,000	$\$140,000 - \$84,000 = \$56,000$	$\$420,000 - \$84,000 \times 4 = \$84,000$
Year	EVA Charge	Annual EVA		PV of Annual EVAs
1	$\$420,000 \times 10\% = \$42,000$	$\$56,000 - \$42,000 = \$14,000$		$\$14,000 \div (1 + 10\%)^1 = \$12,727$
2	$\$336,000 \times 10\% = \$33,600$	$\$56,000 - \$33,600 = \$22,400$		$\$22,400 \div (1 + 10\%)^2 = \$18,512$
3	$\$252,000 \times 10\% = \$25,200$	$\$56,000 - \$25,200 = \$30,800$		$\$30,800 \div (1 + 10\%)^3 = \$23,140$
4	$\$168,000 \times 10\% = \$16,800$	$\$56,000 - \$16,800 = \$39,200$		$\$39,200 \div (1 + 10\%)^4 = \$26,774$
5	$\$84,000 \times 10\% = \$8,400$	$\$56,000 - \$8,400 = \$47,600$		$\$47,600 \div (1 + 10\%)^5 = \underline{\underline{\$29,556}}$
Sum PVs to find Project EVA = \$110,710				

Economic Value Added (EVA)

Project B				
Year	OCF	Depreciation	NOPAT	Invested Capital
1	\$280,000	\$90,000	$\$280,000 - \$90,000 = \$190,000$	$\$450,000 - \$90,000 \times 0 = \$450,000$
2	\$120,000	\$90,000	$\$120,000 - \$90,000 = \$30,000$	$\$450,000 - \$90,000 \times 1 = \$360,000$
3	\$100,000	\$90,000	$\$100,000 - \$90,000 = \$10,000$	$\$450,000 - \$90,000 \times 2 = \$270,000$
4	\$100,000	\$90,000	$\$100,000 - \$90,000 = \$10,000$	$\$450,000 - \$90,000 \times 3 = \$180,000$
5	\$100,000	\$90,000	$\$100,000 - \$90,000 = \$10,000$	$\$450,000 - \$90,000 \times 4 = \$90,000$
	EVA Charge	Annual EVA		PV of Annual EVAs
1	$\$450,000 \times 10\% = \$45,000$	$\$190,000 - \$45,000 = \$145,000$		$\$145,000 \div (1 + 10\%)^1 = \$131,818$
2	$\$360,000 \times 10\% = \$36,000$	$\$30,000 - \$36,000 = -\$6,000$		$-\$6,000 \div (1 + 10\%)^2 = -\$4,959$
3	$\$270,000 \times 10\% = \$27,000$	$\$10,000 - \$27,000 = -\$17,000$		$-\$17,000 \div (1 + 10\%)^3 = -\$12,772$
4	$\$180,000 \times 10\% = \$18,000$	$\$10,000 - \$18,000 = -\$8,000$		$-\$8,000 \div (1 + 10\%)^4 = -\$5,464$
5	$\$90,000 \times 10\% = \$9,000$	$\$10,000 - \$9,000 = \$1,000$		$\$1,000 \div (1 + 10\%)^5 = \621
Sum PVs to find Project EVA =				\$109,244

The CAPM: Project-Specific and Group-Specific Required Rates of Return

- Capital-Asset Pricing Model Approach to Project Selection
- *Use of CAPM in Project Selection*
 - Initially assume all-equity financing.
 - Determine project beta.
 - Calculate the expected return.
 - Adjust for capital structure of firm.
 - Compare cost to IRR of project.

The CAPM: Project-Specific and Group-Specific Required Rates of Return

- Capital-Asset Pricing Model Approach to Project Selection
- For such a firm, the CAPM approach to determining a required return is equivalent to determining the cost of equity capital of a firm.
- However, instead of the expected relationship between excess returns for common stock (returns in excess of the risk-free rate) and those for the market portfolio, one is concerned with the expected relationship of excess returns for a project and those for the market portfolio.
- The required return for an equity-financed project, therefore, would be

The CAPM: Project-Specific and Group-Specific Required Rates of Return

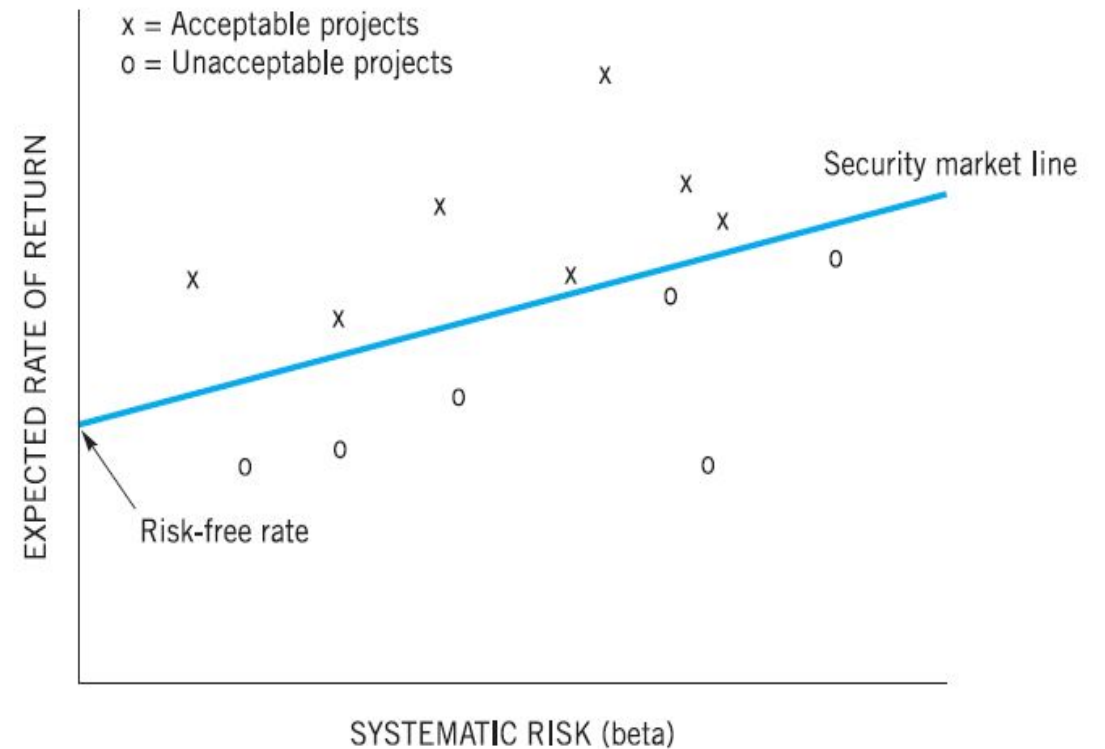
- The required return for an equity-financed project, therefore, would be

$$R_k = R_f + (\bar{R}_m - R_f)\beta_k$$

- where β_k is the slope of the *characteristic line* that describes the relationship between excess returns for project k and those for the market portfolio.
- Assuming that the firm intends to finance a project entirely with equity, the acceptance criterion would then be to invest in the project if its expected return met or exceeded the required return, R_k ,

The CAPM: Project-Specific and Group-Specific Required Rates of Return

- The line in the figure represents the security market line – the market-determined relationship between required rate of return and systematic risk.
- All projects with internal rates of return lying on or above the line should be accepted, because they are expected to provide returns greater than or equal to their respective required returns.
- Acceptable projects are depicted by x's. All projects lying below the line, shown by the o's, would be rejected. Note that the greater the systematic risk of a project, the greater the return that is required.
- If a project had no systematic risk, only the risk-free rate would be required. For projects with more risk, however, a risk premium is demanded, and it increases with the degree of systematic risk of the project.



The CAPM: Project-Specific and Group-Specific Required Rates of Return

- Application of the Model – the Use of Proxy Companies.
- The difficulty in applying the CAPM approach to project selection is in estimating the beta for a project.
- It is therefore necessary to estimate a project's period returns in terms of its periodic cash flows and its change in value from the beginning of each period to the end of that period.
- The difficulty is in estimating the value of the project from period to period. Unfortunately, the values of assets that are not publicly traded are not directly observable, so we cannot calculate beta in the manner presented earlier for a publicly traded stock.
- However, in many cases a project is sufficiently similar to a company whose stock is publicly held so that we can use that company's beta in deriving the required rate of return on equity for the project. The important thing is to identify a company or companies with systematic risk characteristics similar to those of the project in question.

The CAPM: Project-Specific and Group-Specific Required Rates of Return

- The Required Return with Leverage.
- If the firm consistently finances its projects only with equity, we would use the required return on equity, R_k , as the required rate of return for the project.
- If some debt financing is employed, however, we need to determine a weighted average required return.
- Rather than vary the proportion of debt financing project by project, a more consistent approach is to apply the same weights to all projects. Presumably these weights will correspond to the proportions with which the firm intends to finance over time.
- If, for example, the firm intends to finance with one part debt for every three parts equity, and the after-tax cost of debt is 6.60 percent and the required return on equity is 16 percent, the overall (weighted average) required return for the project is

$$\begin{aligned}\text{Weighted average required return} &= \left[\begin{array}{c} \text{Cost of} \\ \text{debt} \end{array} \right] \left[\begin{array}{c} \text{Proportion} \\ \text{of debt} \end{array} \right] + \left[\begin{array}{c} \text{Cost of} \\ \text{equity} \end{array} \right] \left[\begin{array}{c} \text{Proportion} \\ \text{of equity} \end{array} \right] \\ &= (0.066)(0.25) + (0.16)(0.75) = \mathbf{13.65\%}\end{aligned}$$

The CAPM: Project-Specific and Group-Specific Required Rates of Return

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- If the project were expected to provide an internal rate of return greater than or equal to this rate, the project would be accepted. If not, it would be rejected. Thus, even for a leveraged firm, the acceptance criterion remains specifically related to the systematic risk of the project through the cost of equity capital.

Group-Specific Required Return

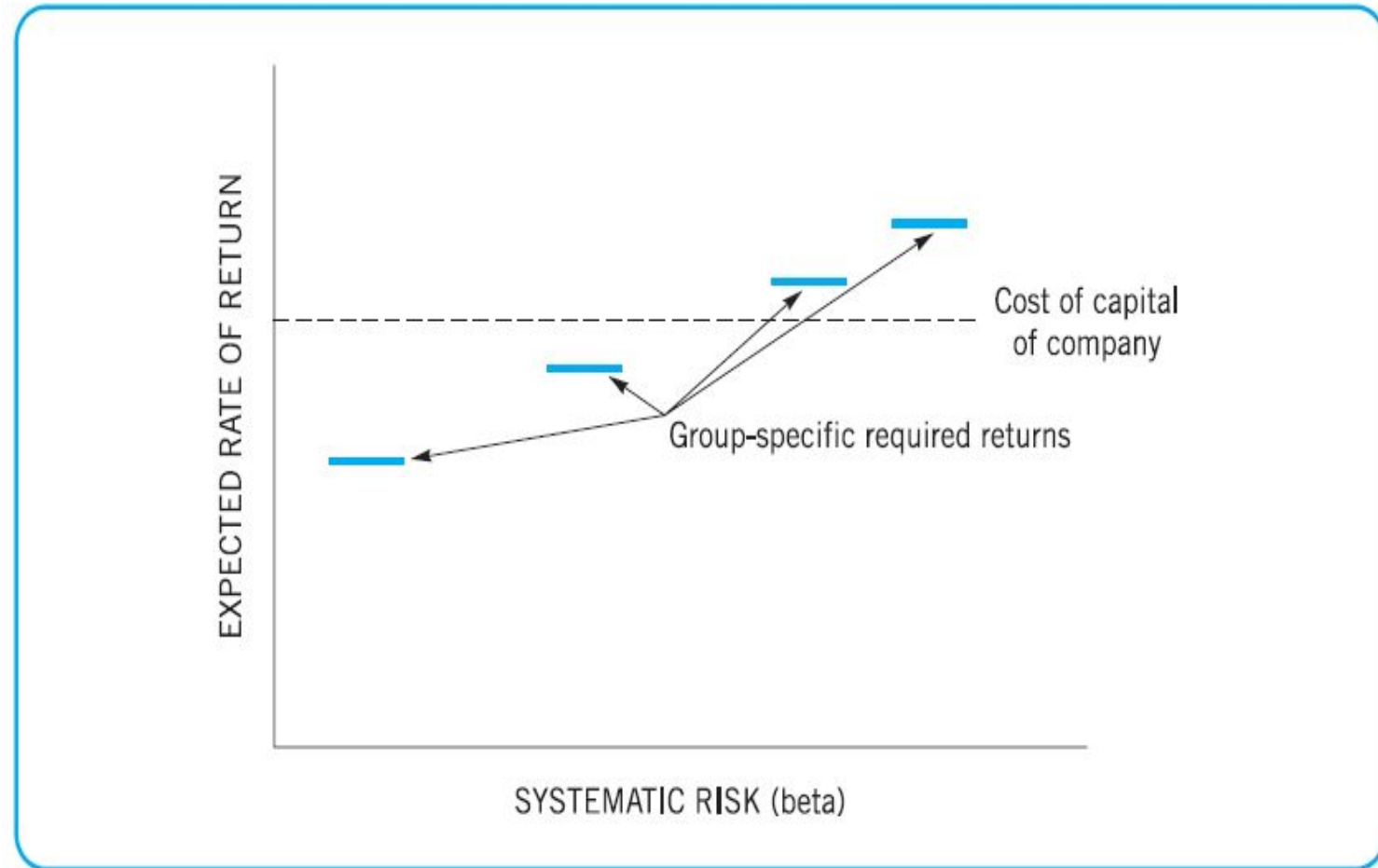
- *Use of CAPM in Project Selection:*
- Initially assume all-equity financing.
- Determine group beta.
- Calculate the expected return.
- Adjust for capital structure of group.
- Compare cost to IRR of group project.

Group-Specific Required Return

- *Use of CAPM in Project Selection:*
- Rather than determine project-specific required returns, some companies categorize projects into roughly risk-equivalent groups and then apply the same CAPM-determined required return to all projects included within that group.
- One advantage of this procedure is that it is not as time-consuming as computing required returns for each project. Another advantage is that it is often easier to find proxy companies for a group than it is for individual projects. By “group,” we mean some subunit of the company that carries on a set of activities that can be differentiated from the other activities of the firm.
- Usually these activities are differentiated along product or service lines as well as along management lines. Frequently the subunits are divisions or **subsidiaries** of the company.
- If the products or services of the group are similar with respect to risk and new proposals are of the same sort, a group-specific required return is an appropriate acceptance criterion. It represents the rate charged by the company against a group for capital employed. Stated differently, it is the rate of return the company expects the group to earn on its capital investments. The greater the systematic risk of the group, the greater its required return.

Group-Specific Required Return

- The computation of the required rate of return is the same as that for the specific project.
- Projects from a group that provide expected returns above their group-specific bar should be accepted.
- Capital is allocated on a risk-return basis specific to the systematic risk of the group

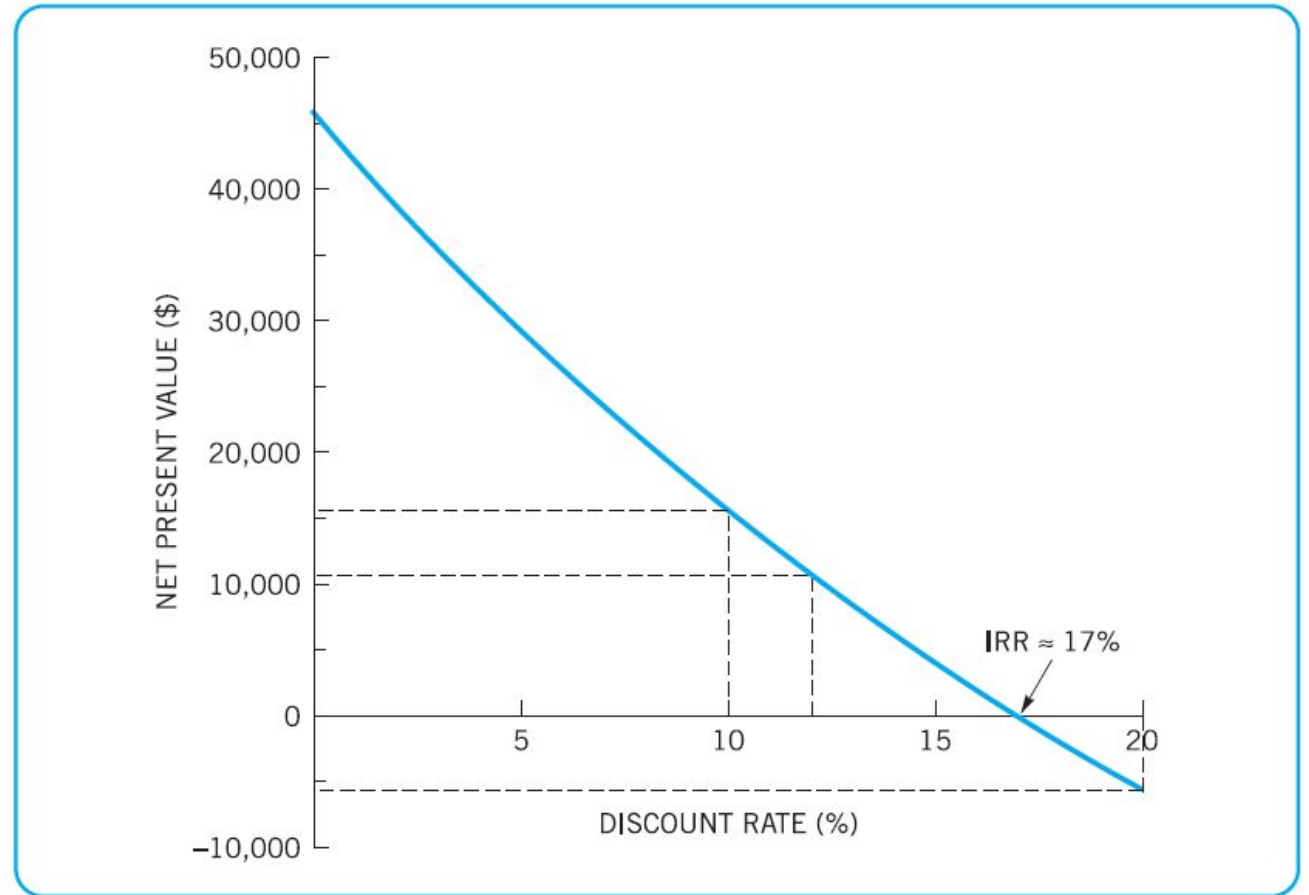


Evaluation of Projects on the Basis of Their Total Risk

- Risk-Adjusted Discount Rate Approach
- For investment proposals having risk similar to that of an “average” project for the firm, we have already seen that the firm’s overall cost of capital can serve as the required rate of return.
- Risk-Adjusted Discount Rate is required return is increased (decreased) relative to the firm’s overall cost of capital for projects or groups showing greater (smaller) than “average” risk.
- The **risk-adjusted discount rate (RADR)** approach to investment proposal selection pertains to projects or groups whose outcomes are considered to have greater or less risk than the “average” undertaking of the firm.

Risk-Adjusted Discount Rate Approach and NPV

- in the RADR method, the discount rate is “adjusted” for risk by increasing it relative to the overall cost of capital to compensate for greater risk and lowering it to account for less risk.
- The project- or group-specific required return then becomes the risk-adjusted discount rate.



Risk-Adjusted Discount Rate Approach and NPV

The RADR approach calls for

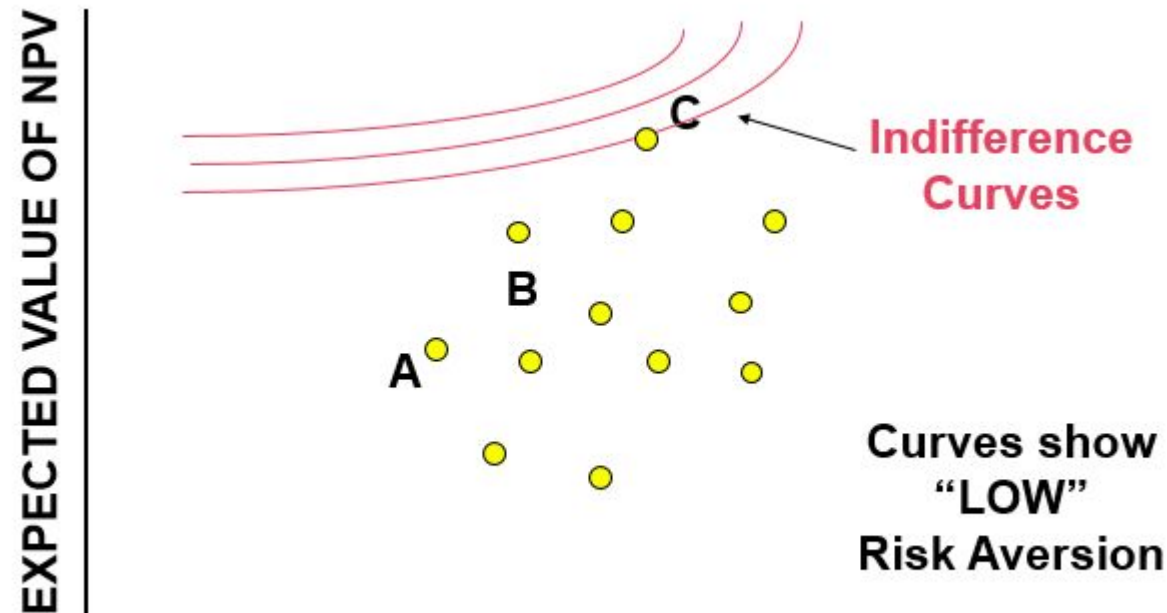
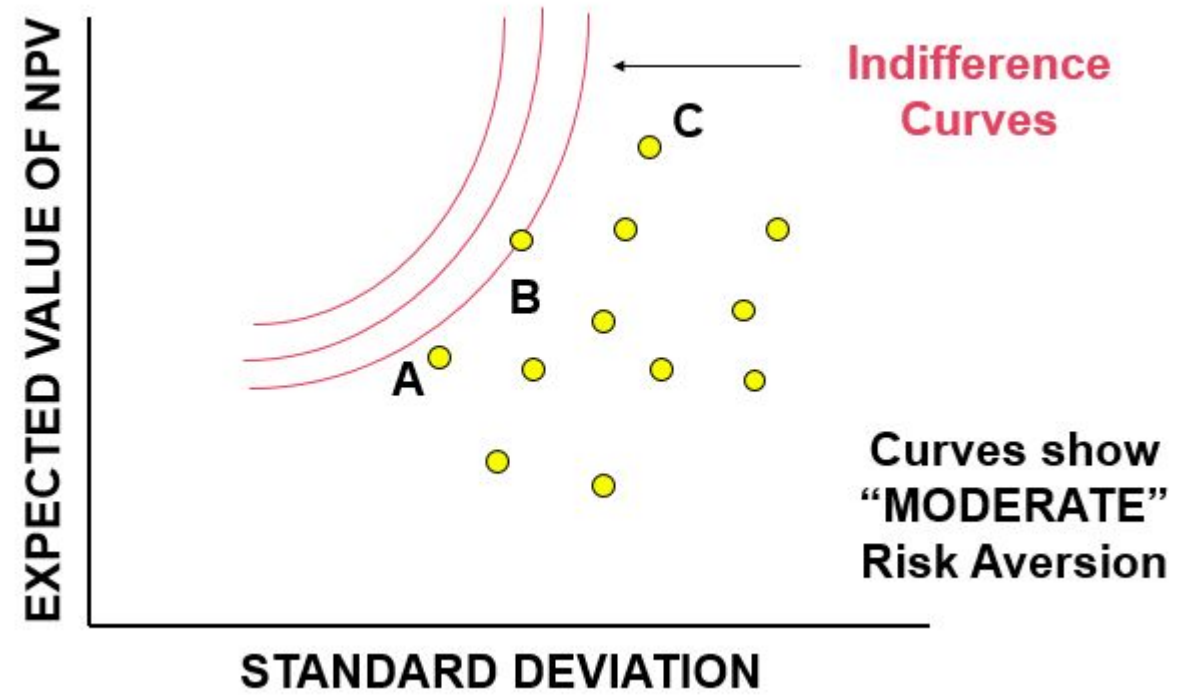
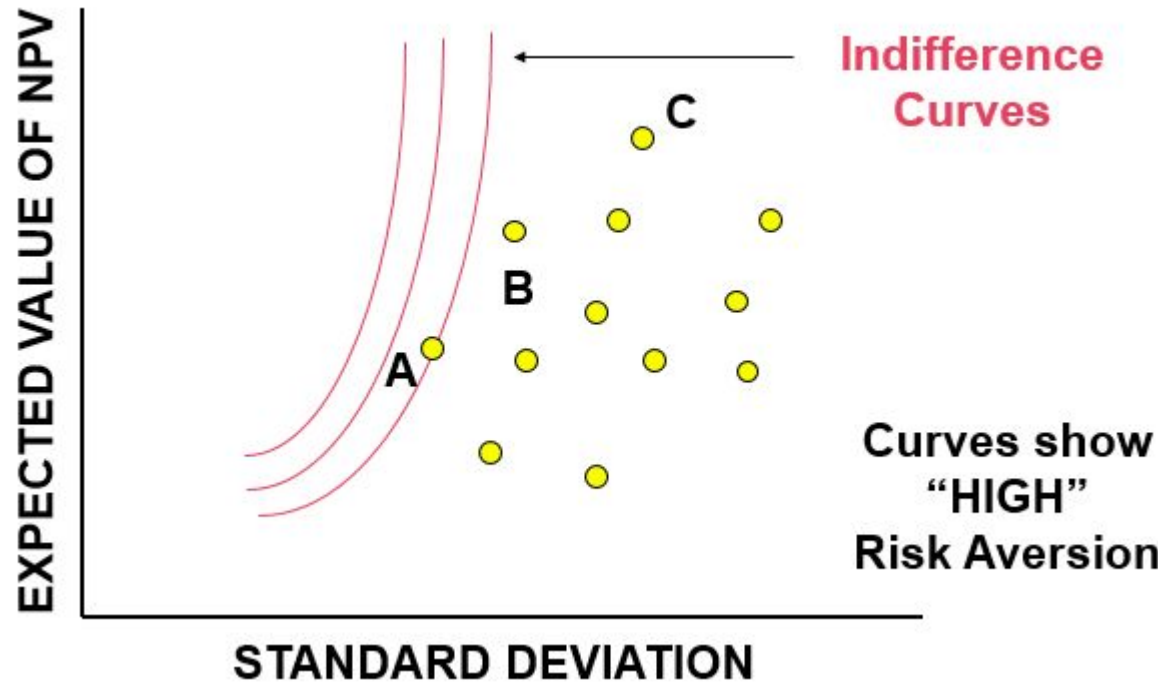
- Adjusting the required return (discount rate) *upward* from the firm's overall cost of capital for projects or groups showing *greater* than “average” risk and
- Adjusting the required return (discount rate) *downward* from the firm's overall cost of capital for projects or groups showing *less* than “average” risk.

Probability Distribution Approach

- Acceptance of a single project with a positive NPV depends on the dispersion of NPVs and the utility preferences of management.

Contribution to Total Firm Risk: Firm-Portfolio Approach

- The appropriate information is the expected value and standard deviation of the probability distribution of possible net present values for all feasible combinations of existing projects and investment proposals under consideration.
- Assume for now that management is interested only in the marginal impact of an investment proposal on the risk complexion of the firm as a whole.
- The selection of the most desirable combination of investments will depend on management's risk preferences with respect to expected net present value and standard deviation.
- Indifference curve A line representing all combinations of expected return and risk that provide an investor with an equal amount of satisfaction.



Adjusting the Beta for Financial Leverage

- In attempting to indirectly calculate a project's (or group's) cost of equity capital, we need to use a proxy company that has business risk similar to that of our project.
- Unfortunately, the proxy firm may use a significantly different proportion of debt than that used by our firm. Therefore it may be necessary to adjust the beta of the proxy company for this difference in capital structure.

The required rate of return for common stock in a firm with financial leverage (a “levered” firm) is

$$R_j = R_f + (\bar{R}_m - R_f)\beta_j \quad [15A.1]$$

where R_f = risk-free rate

$\bar{R}_m$ = expected return on the market portfolio

β_j = beta measuring the systematic risk of the equity of the levered firm

Adjusting the Beta for Financial Leverage

Equation (15A.1) can also be expressed as¹⁴

$$R_j = R_f + (\bar{R}_m - R_f)\beta_{ju}[1 + (B/S)(1 - T_c)] \quad (15A.2)$$

where β_{ju} = beta measuring the systematic risk of the equity of the firm in the *absence of financial leverage* (the beta if the firm were all-equity-financed or “unlevered”)

B/S = debt-to-equity ratio in market value terms

T_c = corporate tax rate

Rewriting Eq. (15A.2), we can clearly see that the required rate of return on equity is composed of the risk-free rate, plus a premium for business risk and a premium for financial risk:

Required return	=	Risk-free rate	+	Business-risk premium	+	Financial-risk premium
R_j	=	R_f	+	$(\bar{R}_m - R_f)\beta_{ju}$	+	$(\bar{R}_m - R_f)\beta_{ju}(B/S)(1 - T_c)$

The measured beta for the stock, β_j , embodies both risks and is simply

$$\beta_j = \beta_{ju}[1 + (B/S)(1 - T_c)] \quad (15A.3)$$

Adjusted Present Value

- **Adjusted Present Value (APV)** is the sum of the discounted value of a project's operating cash flows plus the value of any tax-shield benefits of interest associated with the project's financing minus any flotation costs.

$$APV = \text{Unlevered project value} + \text{Value of project financing}$$

- **Tax shield:** A tax deductible expense. The expense protects (shields) an equivalent dollar amount of revenue from being taxed by reducing taxable income.

$$APV = \left[\sum_{t=1}^n \frac{CF_t}{(1 + k_{eu})^t} - ICO \right] + \left[\sum_{t=1}^n \frac{(I_t)(T_c)}{(1 + k_d)^t} - F \right]$$

Adjusted Present Value

Assume *Basket Wonders* is considering a new \$425,000 automated basket weaving machine that will save \$100,000 per year for the next 6 years. The required rate on unlevered equity is 11%.

BW can borrow \$180,000 at 7% with \$10,000 after-tax flotation costs. Principal is repaid at \$30,000 per year (+ interest). The firm is in the 40% tax bracket.

- What is the NPV to an all-equity-financed firm?

$$NPV = \$100,000[PVIFA_{11\%,6}] - \$425,000$$

$$NPV = \$423,054 - \$425,000$$

$$NPV = -\$1,946$$

What is the *APV*?

First, determine the interest expense.

<u>Int Yr 1</u>	$(\$180,000)(7\%)$	$= \$12,600$
<u>Int Yr 2</u>	$(\$150,000)(7\%)$	$= 10,500$
<u>Int Yr 3</u>	$(\$120,000)(7\%)$	$= 8,400$
<u>Int Yr 4</u>	$(\$90,000)(7\%)$	$= 6,300$
<u>Int Yr 5</u>	$(\$60,000)(7\%)$	$= 4,200$
<u>Int Yr 6</u>	$(\$30,000)(7\%)$	$= 2,100$

Second, calculate the tax-shield benefits.

TSB Yr 1	$(\$12,600)(40\%)$	$= \$5,040$
TSB Yr 2	$(\$10,500)(40\%)$	$= 4,200$
TSB Yr 3	$(\$8,400)(40\%)$	$= 3,360$
TSB Yr 4	$(\$6,300)(40\%)$	$= 2,520$
TSB Yr 5	$(\$4,200)(40\%)$	$= 1,680$
TSB Yr 6	$(\$2,100)(40\%)$	$= 840$

Third, find the PV of the tax-shield benefits

TSB Yr 1	$(\$5,040)(.901)$	$= \$4,541$
TSB Yr 2	$(\$4,200)(.812)$	$= 3,410$
TSB Yr 3	$(\$3,360)(.731)$	$= 2,456$
TSB Yr 4	$(\$2,520)(.659)$	$= 1,661$
TSB Yr 5	$(\$1,680)(.593)$	$= 996$
TSB Yr 6	$(\$840)(.535)$	$= 449$

PV = \$13,513

What is the *APV*?

$$APV = NPV + PV \text{ of TS} - \text{Flotation Cost}$$

$$APV = -\$1,946 + \$13,513 - \$10,000$$

$$APV = \$1,567$$